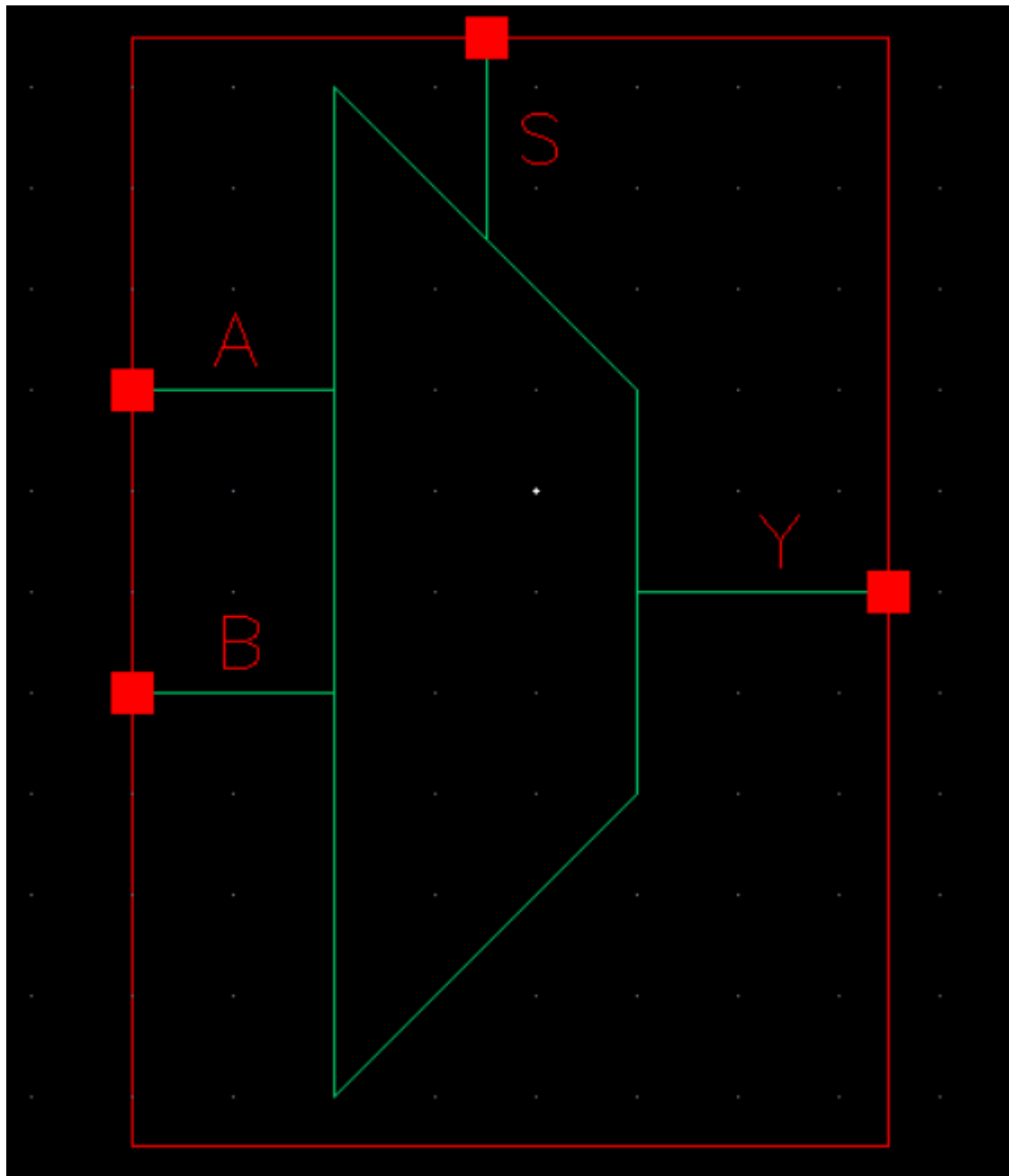


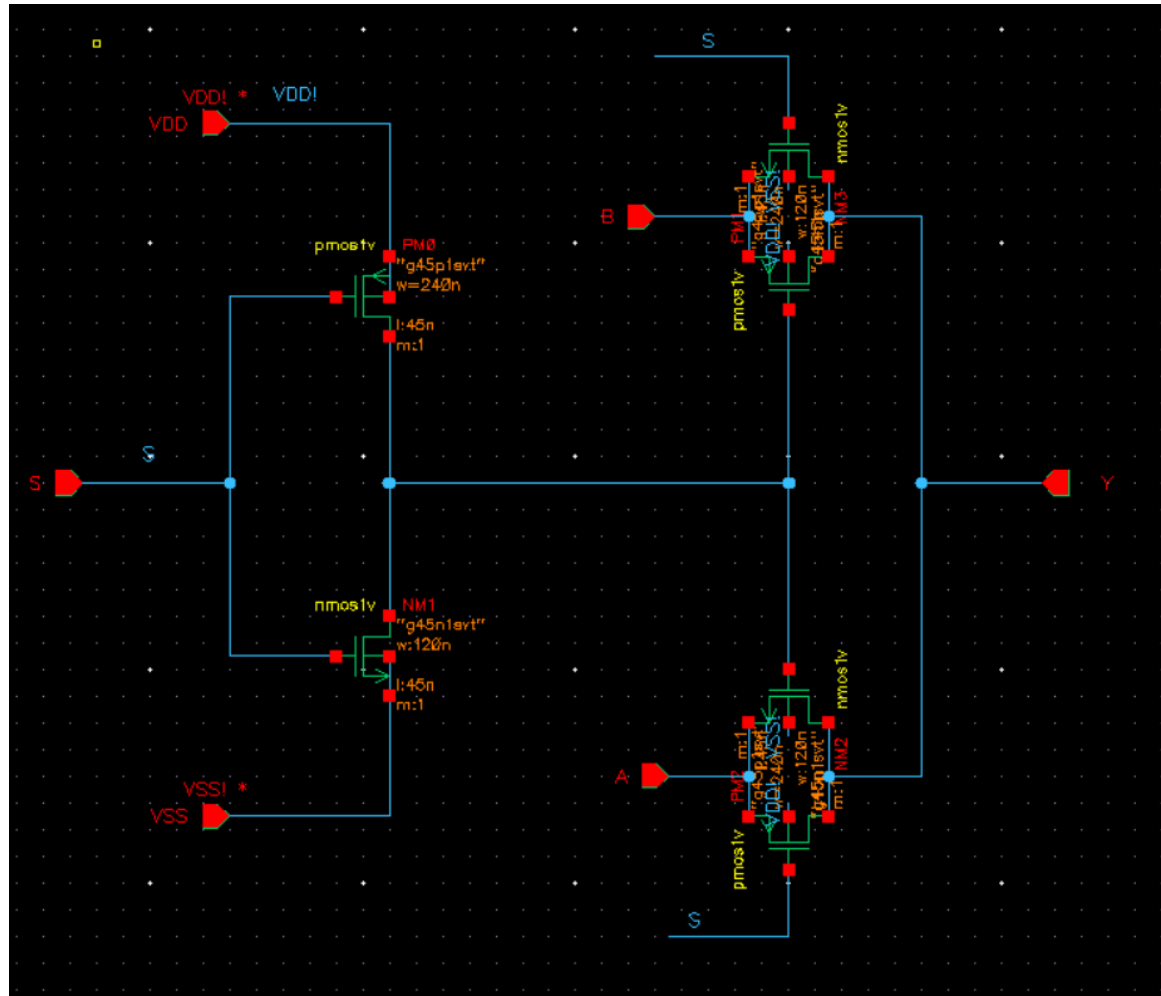
Library Name:		cgm	
Cell Name:		MUX2XI	
Function/Truth Table:			
S	A	B	Y
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1
Propagation Delay (path based): with extracted parasitics			
Transition			Delay (s)
A → Y ↑ (T_PLH)			959.7p
A → Y ↓ (T_PHL)			1.127n
B → Y ↑ (T_PLH)			961.2P
B → Y ↓ (T_PHL)			1.126n
S → Y ↑ (T_PLH)			998.4p
S → Y ↓ (T_PHL)			1.163n
Output Rise Time (path based):			
Transition			Rise Time (s)

A → Y ↑	2.297n
B → Y ↑	2.309n
S → Y ↑	2.277n
Output Fall Time (path based):	
Transition	Fall Time (s)
A → Y ↓	2.841n
B → Y ↓	2.849n
S → Y ↓	2.84n
Layout Height: 1.71μm	
Layout Width: 1.6μm	
Layout Area: 2.726μm²	

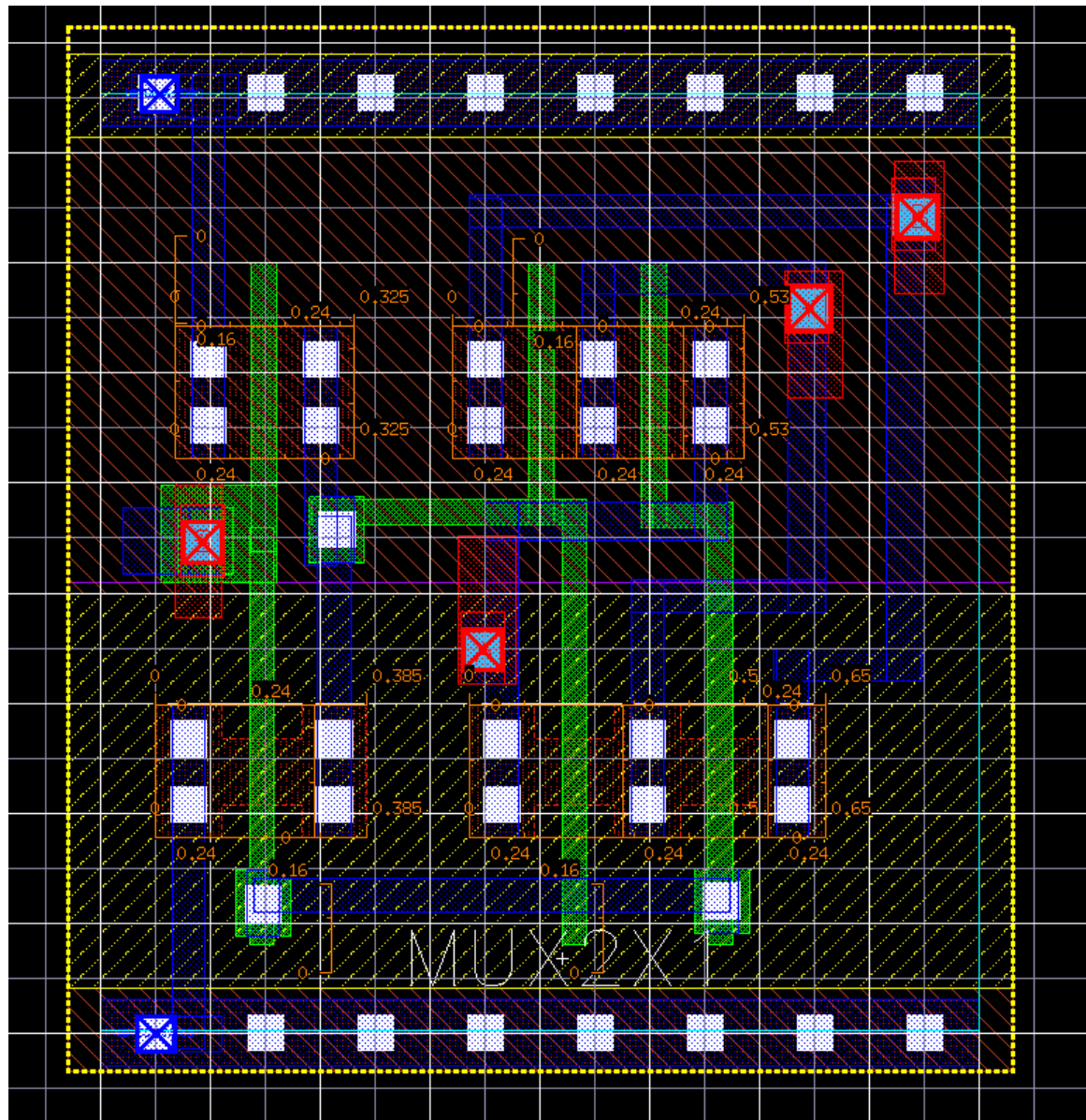
Symbol with Port Names:



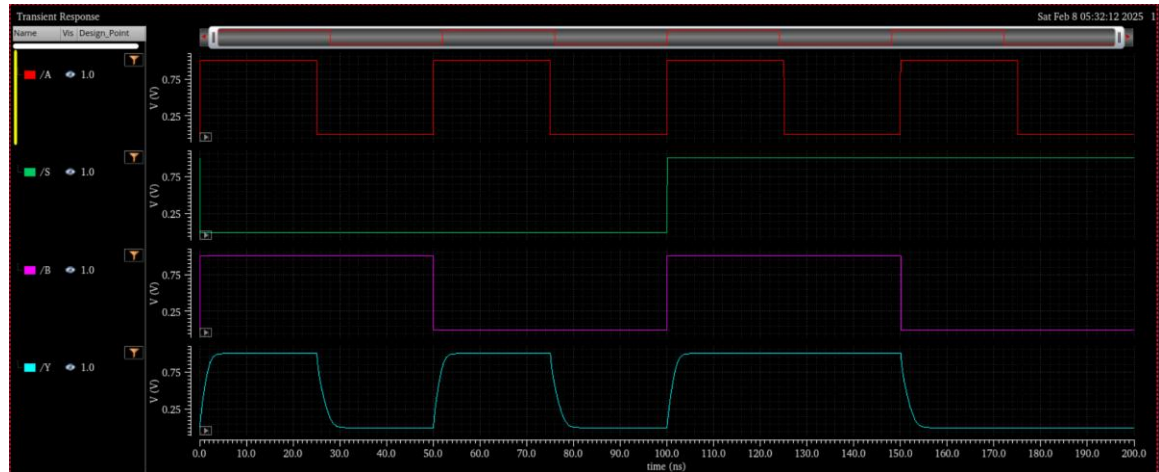
Schematic:



Layout:



Functional Simulation Waveforms



Comments/Notes:

The MUX2X1 gate was tested and verified. Y follows A when $S = 0$ and B when $S = 1$, confirming correct functionality. Propagation delays and rise/fall times were within expected ranges, with fall transitions being slightly longer. All functional and timing tests passed, ensuring reliable operation.